Deep vein thrombosis | DVT or poketoto

Blood clots in deep veins

What is a deep vein thrombosis?

A deep vein thrombosis (DVT) is a blood clot (or thrombus) that forms in veins that are deep under your skin – usually within your muscles. Veins are the blood vessels that carry blood from the tissues of your body, such as your muscles, liver, brain and gut, back to your heart. The most common veins to be affected in DVT are those in your legs.

The blood clot can travel through your veins to another part of your body. This is called an embolus. If the embolus travels and gets stuck in your lungs, it's called a **pulmonary embolism** – this is the most serious complication of a DVT.

What causes a DVT?

Sometimes a DVT happens for no known reason, but there are things that can increase your risk of having a DVT.

- When blood flow in your veins is slower than normal and more likely to clot, such as if you spend a long time not moving.
- When there's damage to the inside surface of your vein, such as from a previous DVT.
- You have a condition or take a medicine that makes your blood clot more easily than normal.

The more risk factors you have, the greater your risk.

A DVT is more likely to happen if you:

- have had a DVT or **pulmonary embolism (PE)** before
- have a family history of DVT or PE
- can't move around much for long periods, such as after a **stroke** or major surgery, or during long-distance travel
- have had recent trauma to a limb or recently had your leg or foot in a cast
- have cancer or **heart failure**
- are pregnant or have had a baby in the previous 6 weeks
- are over 60 years of age
- smoke
- take the **contraceptive pill** or **menopausal hormone therapy** (previously called hormone replacement therapy or HRT)
- have thrombophilia, a blood disorder that increases your risk of developing abnormal blood clots in your veins or arteries
- are overweight
- have **varicose veins**.
- are **dehydrated**.

What are the symptoms of DVT?

You can get a DVT in any limb (arm or leg) but they tend to occur more commonly in your legs.

Image credit: 123rf

Signs and symptoms in the affected limb can include:

- Swelling
- pain – in your leg the pain often starts in your calf and can feel like cramping or a soreness.
- change of skin colour – your skin may become bluish, purple or reddish
- a feeling of warmth on the skin over the painful or swollen area.

These symptoms may often be confused with a sprained ankle or other injury. Sometimes a DVT can happen without any noticeable symptoms.

If you develop signs or symptoms of DVT, contact your healthcare provider urgently.

Seek immediate medical help if you have symptoms of DVT, such as pain and swelling, and any of the following:

- breathlessness
- chest pain
- coughing up blood
- fainting or collapse.

These may be signs that a blood clot has travelled to your lungs (pulmonary embolism). **A pulmonary embolism can be life threatening and needs treatment straight away.**

Call Healthline on 0800 611 116 if you're not sure what to do.

How is a DVT diagnosed?

If you suspect you have a DVT, even if you have mild symptoms, you must seek medical attention urgently. This can prevent serious complications such as a **pulmonary embolism.**

To diagnose DVT, your healthcare provider will ask you questions about your symptoms and any risk factors you may have.

They'll also examine you to check for areas of swelling, tenderness or colour change on your skin.

Your healthcare provider may suggest further testing, such as a **D-dimer blood test** and/or an **ultrasound scan**. A D-dimer blood test helps show whether a blood clot is likely.

- If your risk of DVT is low and your D-dimer test is negative, you may not need any other tests.
- If your D-dimer test is positive, or your symptoms suggest a higher chance of DVT, an ultrasound scan is usually needed.

How is a DVT treated?

The main aim of DVT treatment is to prevent the clot from getting any bigger and from breaking loose and causing a pulmonary embolism. After that, the goal is to reduce your chances of DVT happening again. Treatment will depend on where your clot is and how bad it is. The pain and swelling usually improve within a few days of starting treatment.

Medicines – anticoagulants

Anticoagulants work by interrupting the clot-forming process. For the treatment of a DVT, anticoagulants can stop the existing clot from getting bigger.

Most people with a DVT are now treated with a direct oral anticoagulant (DOAC), such as **rivaroxaban** or **dabigatran**. These medicines start working quickly and you don't usually need regular blood test monitoring. You may need a different anticoagulant depending on if you're pregnant (hapū), your kidney function, bleeding risk or other medical conditions.

How long you'll need anticoagulation treatment depends on many factors, but it's usually taken for at least 3 months. You may need it for longer, even for the rest of your life.

A DVT in pregnancy is treated differently and usually needs a longer course of treatment. Your healthcare provider will advise you about the treatments available.

While taking anticoagulants, you need to watch for signs of bleeding. These medicines help to prevent dangerous blood clots from forming or getting bigger, but they can also make bleeding more of a problem. You should:

- take the medicine exactly as prescribed
- tell health professionals and dentists you're taking an anticoagulant
- check with a pharmacist before taking new medicines
- avoid anti-inflammatory pain medicines such as ibuprofen unless advised by your healthcare provider as these increase your risk of bleeding
- avoid situations where you might get injured
- seek medical advice if you have unusual bruising, bleeding that goes on for a long time, blood in your pee or poo, or severe headaches.

You may like to consider getting a **MedicAlert** bracelet or necklace. If you need treatment but you're too unwell to explain your medical conditions, this lets medical personnel know that you're taking an anticoagulant and are at risk of bleeding.

Compression stockings

These stockings, also called 'graded compression stockings', are made from special elastic to give support to your lower legs. They improve your blood circulation and help to reduce swelling.

Some people may be advised to wear compression stockings to help reduce swelling and discomfort after a DVT but they're no longer recommended for everyone with a DVT. Your healthcare provider will tell you if they're likely to help in your situation.

Read more about **compression stockings**.

What self-care can I do with a DVT?

If you have a DVT you should:

- take any anticoagulant medicine that's been prescribed for you exactly as directed
- stay as active as your healthcare provider recommends
- avoid sitting or standing for long periods
- raise your leg when resting to help reduce swelling
- wear compression stockings if they've been prescribed
- stay well hydrated
- go to any follow-up appointments.

Seek emergency medical care immediately if you develop sudden shortness of breath, chest pain, coughing up blood, or you faint, as these could be signs that the clot has travelled to your lungs.

Image credit: Healthify He Puna Waiora

How can a DVT be prevented? (+ APPS) (+ VIDEO)

A DVT can't always be prevented, but there are a few things you can do to lower your risk:

- Avoid sitting for long periods of time. Elevate (raise) your legs if you do sit for a long time.
- Don't cross your legs when you're sitting down.
- Get up and move around every hour or so.
- If you are travelling on a long journey or you're confined to bed in hospital there are special precautions you can take. See **travel-related DVT** and **hospital-related DVT** below.
- If you smoke, aim to quit or reduce the amount that you smoke. Smoking increases the risk of blood clots and many other health conditions. Stopping smoking can improve your overall health and reduce your risk of DVT.
- If you're overweight, aim to reduce weight.
- Stay physically active.
- Keep hydrated by drinking plenty of fluids – especially when you're travelling or unwell.
- Avoid alcohol and medications that make you drowsy because they make it harder for you to move around.

Video: Deep vein thrombosis prevention exercises

(AskDoctorJo, US, 2016)

Apps reviewed by Healthify

You may find it useful to look at some **nutrition, exercise and weight management app** and **quit smoking apps**.

Travel-related DVT

If you're travelling for more than 4 hours by air, car, bus or train, you can be at risk for blood clots. This is because when you're sitting for long periods your blood flows more slowly. The slower blood flow means that blood can collect in your legs and cause a blood clot (DVT).

For most people, the chance of developing a DVT just from taking a long journey is very small.

If you develop a swollen, painful, red, lower leg or breathing difficulties shortly after a long journey then see a healthcare provider urgently.

Who is at risk of travel-related DVT?

Anyone may develop a DVT following a long journey, but the risk is increased for some groups – see 'Who's at risk of developing DVT?' above. See your healthcare provider before travelling on a long journey so you can discuss your risk and what action you may need to take.

Tips to reduce your risk of travel-related DVT

Most people have no problems during travel – many people get slight, painless puffiness of feet and ankles after a long journey but this isn't due to a DVT.

When travelling on a long journey, particularly on a long-haul plane trip, here are a few things you can do to reduce your risk of DVT.

Do simple exercises

- Exercise your calf and foot muscles regularly. Bend and straighten your legs, feet and toes when you're seated. Repeat this every half hour or so.
- Press the balls of your feet down hard against the floor or foot rest every so often. This helps to increase the blood flow in your legs.
- If you're on a plane, walk up and down the aisle every hour or so, when the seatbelt signs aren't switched on.
- Get up to stretch your legs, when there are stops in your journey.

Drink plenty of water

- Drink liquids, preferably water, regularly to avoid **dehydration**.
- Don't drink too much alcohol. Alcohol can cause dehydration and difficulty moving around.

Avoid sleeping tablets

- Don't take sleeping tablets as they can mean you stay in the same position for a long time which increases the risk of DVT.

Wear compression stockings

Correctly fitted compression stockings can prevent DVT if you're at increased risk. The slight pressure from the stocking helps to prevent blood ‘pooling’ in your lower legs.

- You can buy the stockings (or 'flight socks') from your pharmacy. Ask the pharmacist for advice about the correct sort.
- Stockings don’t replace the need for regular exercises. They should be worn in addition to doing exercises.
- If you have any of the risk factors mentioned above, it would be wise to wear these stockings for your journey.

Read more about **compression stockings**.

Medicine

Medicine to prevent DVT when travelling for long distances isn’t usually recommended. It’s more important to move around often and avoid dehydration, alcohol and sleeping tablets. However, if you’re at high risk of having a DVT, speak to your healthcare provider about whether medicine to prevent DVT is suitable for you.

After the journey

Take a little walk straight after the journey to get your circulation going. Most travellers have no problems – many people get slight, painless puffiness of their feet and ankles after a long journey, but this isn’t due to a DVT.

If you develop a swollen painful, red lower leg or breathing difficulties shortly after a long journey then see a healthcare provider urgently.

Hospital-related DVT

Usually, muscle movements help to push the blood around your body by squeezing the veins. This helps blood to flow freely without clotting. If you are not mobile for some time, blood flow can become slow and allow a clot to form. Read more about **blood clots.**

If you're in hospital after surgery or an injury, and as your body tries to protect itself from blood loss, it causes your blood to clot more easily. However, this also increases the risk of unwanted clotting such as a DVT.

Who is most at risk?

The risk of DVT is highest in people who have had:

- major joint surgery (eg, hip or knee replacement)
- major surgery to the abdomen (tummy), hips, chest or legs
- major surgery for cancer
- severe physical injury or trauma
- serious sudden illness such as **heart attack**, **heart failure**, **stroke**, **spinal cord injury**, severe infection, cancer or **chronic obstructive pulmonary disease** (COPD).

If you're at risk of developing DVT, there are things that can be done to help prevent a blood clot forming.

Before going to hospital

If you're going to hospital to have an operation, your healthcare provider will assess your risk of DVT.

- If you're taking medicines that could increase your risk of getting a clot, you may be advised to stop taking your medicine a few weeks before your operation. Examples include:
  - **combined oral contraceptive pill** ('the pill')
  - **menopausal hormone therapy** (MHT)
  - medicines that change the way your blood clots such as **anticoagulants** or **aspirin**.
- Don't stop taking your medicines without speaking to your healthcare provider – they'll tell you which ones, if any, you need to stop.

While you're in hospital

Depending on your risk factors and individual circumstances, your healthcare team will discuss treatment options with you. Treatment may include:

- wearing **compression stockings**
- using a compression pump on your lower legs
- taking tablets or injections (**anticoagulant medication**) to prevent a blood clot
- gently exercising your feet and legs in bed
- getting out of bed and walking as soon as possible.

If you experience any of the following while you are in hospital, call a nurse immediately:

- pain or swelling in your legs
- pain in your lungs or chest

- difficulty breathing.

When you leave hospital

You may need to continue to take anticoagulant medicine and wear compression stockings when you leave the hospital. Before you leave, your healthcare team should advise you about how to use your treatment, how long to continue using it for, and who to contact if you experience any of the symptoms described above.

What is the outlook if I have a DVT?

Many people who have a DVT or pulmonary embolism (PE) will have another DVT or PE at a later time. The risk depends on what caused the clot. The risk is reduced if you have appropriate anticoagulant treatment.

Information for health providers

Tran HA, Gibbs H, Merriman E, et al. **New Guidelines from the Thrombosis and Haemostasis Society of Australia and NZ for the Diagnosis and Management of Venous Thromboembolism** (https://www.mja.com.au/journal/2019/210/5/new-guidelines-thrombosis-and-haemostasis-society-australia-and-new-zealand) MJA 2019;210(5)

Oral anticoagulant selection in primary care (https://bpac.org.nz/2023/anticoagulants.aspx) BPAC, NZ, 2023 (updated 2025)

MedCase – Managing DVT with rivaroxaban (https://www.goodfellowunit.org/medcases/managing-dvt-rivaroxaban) Goodfellow Unit, NZ, 2024

Scan the QR code to view the full page:
https://healthify.nz/health-a-z/d/deep-vein-thrombosis

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